STUDYNEST | CLASS 11 PHYSICS

QUESTION PAPER 1

Chapter 2: Motion in a Straight Line

Maximum Marks: 40 | Suggested time: 60 minutes

Instructions: Answer all questions. Show calculations and units. Original practice material, not an official examination.

Section A - 1-mark questions - 8 x 1 = 8 marks

1. Define instantaneous velocity. [1]
2. Slope of an x-t graph? [1]
3. Acceleration at constant velocity? [1]
4. Can speed be negative? [1]
5. SI unit of acceleration? [1]
6. Runner returns to start: net displacement? [1]
7. Equation linking v,u,a,t? [1]
8. Area under v-t graph? [1]

Section B - 2-mark questions - 6 x 2 = 12 marks

9. 150 m in 30 s: average speed? [2]
10. Velocity 4 to 16 m/s in 6 s: acceleration? [2]
11. 10 m/s car brakes at 2 m/s^2: stopping time? [2]
12. Convert 90 km/h to m/s. [2]
13. Move 8 m east then 3 m west: distance/displacement? [2]
14. From rest at 3 m/s^2 for 4 s: final speed? [2]

Section C - 3-mark questions - 4 x 3 = 12 marks

15. u=6 m/s, a=2 m/s^2, t=5 s: find v,s. [3]
16. Train at 20 m/s stops in 5 s: a and distance? [3]
17. Velocity rises 2 to 14 m/s in 6 s: a and displacement? [3]
18. u=12 m/s, a=1.5 m/s^2 for 8 s: v,s? [3]

Section D - 4-mark questions - 2 x 4 = 8 marks

19. Car at 25 m/s brakes at 5 m/s^2: stop time and distance? [4]
20. x=4+3t-t^2: v(t), a and time at rest? [4]

End of paper - check working, units, signs and significant figures.